

Eduardo Venegas Hernandez

Data & AI Engineer | Munich, Germany
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SKILLS

AI & Data: Gen AI, PyTorch, LangGraph, LangChain, RAG, PySpark, Vector DB, SQL, Knowledge Graphs, ETL Pipelines

Software & Frameworks: Python, C/C++, TypeScript/JavaScript, FastAPI, Flask, Node.js, React, Next.js, Angular, Streamlit

MLOps & DevOps: MLflow, Docker, Kubernetes, CI/CD, AWS, Azure, Linux, Git, Grafana, MQTT, PLCs

WORK EXPERIENCE

Data & AI Engineer Masterarbeit BMW & DFKI

Munich, Germany 03/2026 - 09/2026

- Topic: Evaluating **Multi-Turn Agentic RAG Systems: A Synthetic Benchmark Generation Approach**.
- **Led the design** of a scalable evaluation framework for precise failure localization in LLM agents.
- Developed a **synthetic data pipeline** using **LangGraph** to benchmark multi-turn conversations.
- Built an **unsupervised auditing framework** to evaluate and scale RAG deployments across **4M+ corporate records**.

AI Research Engineer Werkstudent August-Wilhelm Scheer Institute

Saarbrücken, Germany 10/2024 - 02/2026

- Industrialized a **KG-RAG** system over unstructured technical documents using **LangChain** and **HuggingFace**.
- Developed a **PyTorch Autoencoder** for real-time anomaly detection across distributed devices via Federated Learning.

Big Data Platform Engineer Oracle

Guadalajara, Mexico 11/2023 - 10/2024

- Engineered **CI/CD testing pipelines** and deployment tools for big data services on **Oracle Cloud (OCI)**.
- Designed modular, testable **automation frameworks** to reduce regression test cycles for big data services.

Data Science & Analytics Intern BMW Group

Munich, Germany 03/2023 - 09/2023

- Developed a high-accuracy anomaly detection model to identify vehicle battery replacements from telemetry data.
- Engineered scalable **data pipelines** in **PySpark** to process vehicle telemetry and cloud log databases.
- Optimized database queries and data quality, leading to a 62% increase in analytical service utilization.
- Developed an interactive dashboard to adjust predictive model parameters and visualize system performance.

AI & DevOps Engineer Xira

Mexico City, Mexico 01/2022 - 04/2022

- Managed **distributed DevOps infrastructure**, establishing automated service monitoring and testing workflows.
- Developed a secure **API management platform** integrating Stripe payments using Next.js and Docker.

Production Engineer Fellow Major League Hacking & Meta

New York, USA 06/2021 - 09/2021

- Selected internationally among 120 out of 20,000 applicants. Implemented monitoring with **Prometheus** and **Grafana**.
- Automated DevOps workflows and **CI/CD** pipelines for production deployments on **AWS EC2** using Docker.

Machine Learning & Full Stack Engineer Jiit Technologies

Guadalajara, Mexico 10/2020 - 06/2021

- Designed custom sales forecasting and data pipelines using **Python** (Flask, SciPy, Prophet, and SQL).
- Developed an ERP web platform to manage inventory, logistics, and financial transactions using Angular.

EDUCATION

MSc. Data Science and Artificial Intelligence Universität des Saarlandes

Saarbrücken, Germany 10/2024 - 09/2026

- *Statistics, ML Cybersecurity, NLP, Machine Translation, Quantum AI, Image Processing & Computer Vision.*
- **Thesis (Seminar Grade 1.0):** Evaluating **Multi-Turn Agentic RAG Systems** (In collaboration with **BMW & DFKI**).

BSc. Computer Science Tecnológico de Monterrey

Guadalajara, Mexico 08/2019 - 12/2023

- *ML, Time series, Algorithms & Optimization, Cloud Computing & Distributed, Vehicle User Interfaces, PLC.*

Computer Science Exchange Student Universität Ulm

Ulm, Germany 10/2022 - 09/2023

- Full scholarship recipient from **DAAD**. *Computer Vision II, Autonomous Driving and Assistance Systems, Statistics.*

Technologist Degree in Automation and Robotics CETI Colomos

Guadalajara, Mexico 08/2015 - 06/2019

- Grade: 94.6 / 100. Five times best software engineering GPA. **60% tuition scholarship due to academic merit.**
- *Control Laboratory (I & II), Industrial Robotics, PLCs, Digital Electronics, Controllers Laboratory, Sensor Systems*

PROJECTS

VectorDrive-Route: Autonomous BEV Trajectory Planning

Github

- Built a **PyTorch** pipeline using **nuScenes**, **ResNet** and **depth lifting** to map multi-camera feeds into a **Bird's-Eye-View** grid.
- Designed a temporal **motion planning** head with **drivable-area** and **occupancy decoders** to generate safe paths.

Daten-Mapping-Framework für die AAS

DOI: 10.17560/atp.v68i5.2836

- Developed a mapping framework from **schemas** to **Asset Administration Shell** to achieve interoperability in **Industry 4.0**.
- Designed a dual alignment strategy separating standard **relational tables** from **hierarchical assets**.

See Your World

Github

- Crafted a hackathon award-winning app for the visually impaired, supporting 11 languages.
- Engineered **Image Captioning DL model** with **TensorFlow**, deployed using **Flask**, **Docker**, and **Google Cloud VM**.

CERTIFICATES

- **Mathematics for Data Science Specialization** HSE - 2021
- **Data Science Professional Certificate** Harvard - 2020
- **Python Data Science Professional Certificate** IBM - 2020
- **Deep Learning Professional Certificate** IBM - 2020

LANGUAGES

- **Spanish:** Native
- **English:** C2
- **German:** C1